$i \in \mathcal{I}$. A mechanism $\langle \mathcal{M}, g \rangle$ together with the utility functions $(u_i)_{i\in\mathcal{I}}$ induce a game $\langle \mathcal{M}, g, (u_i)_{i\in\mathcal{I}} \rangle$, where each utility u_i is evaluated at $g(\mu)$ for each agent $i \in \mathcal{I}$. Next, we introduce the notation m_{-i}, i.e., the message profile of all agents except agent $i \in \mathcal{I}$, i.e., $m_{-i} = (m_1, \ldots, m_{i-1}, m_{i+1}, \ldots, m_n)$. Next, consider a game $\langle \mathcal{M}, g, (u_i)_{i\in\mathcal{I}} \rangle$. The solution concept of NE is a message profile μ^* such that $u_i(g(m_i^*, m_{-i}^*)) \geq u_i(g(m_i, m_{-i}^*))$, for all $m_i \in \mathcal{M}_i$ and for each $i \in \mathcal{I}$, where $m_{-i} = (m_1, \ldots, m_{i-1}, m_{i+1}, \ldots, m_n)$. Note that a NE requires complete information. But, we can interpret a NE as the fixed point of an iterative process in an incomplete information setting [S21], [S22]. This is in accordance with John Nash's interpretation of a NE, i.e., the complete information NE can be a possible equilibrium of an iterative learning process.

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Using implementation theory and taking advantage of an optimization-based formulation the focus is to carefully design the outcome function g, i.e., design the payment function p_i such that the allocations $\mathbf{x}$ are efficient for all agents. The standard analysis of such problems is as follows: first, we show that at least one NE exists for the game induced by the mechanism $\langle \mathcal{M}, g \rangle$. Then, since agents are considered to be strategic, we need to ensure that the outcome function (and thus, the payment functions) induce *voluntary participation*, i.e., at any NE, no agent may lose utility from participating in the mechanism. So, in other words, all agents may at the very least have neutral utility ($u_i = 0$). Naturally, in communication networks as well as any other engineering applications we need to be mindful of all monetary transactions in the system. Thus, by checking the sum of all the payments of all the agents, we need to ensure it is zero at NE, which in that case we say the mechanism is *budget balanced*. The next two properties are related to the optimal solution of the optimization problem. Can we ensure that all NE of the induced game $\langle \mathcal{M}, g, (u_i)_{i\in\mathcal{I}} \rangle$ are equivalent to the optimal solution? If yes, then we say that the mechanism fully implements the efficient allocation vector $\mathbf{x}^*$ at a NE. Rarely, we might have an ideal mechanism that results a unique NE. Finally, depending on the network's topology and other physical constraints of our problem of interest, we need to check feasibility, i.e., the allocation $\mathbf{x} = (x_1, \ldots, x_n)$ for each agent is feasible satisfying all the constraints at

NE. This methodology has been widely used in the communication networks literature, and so, in Table 1 we provide a snapshot of recent work that mostly focuses on the utility maximization problem.

In the next subsection, we focus one particularly interesting method that has been studied extensively from economics in communication networks. Of course, other mechanisms have also been investigated in the literature [93]–[97].

TABLE 1: A summary of recent papers on utility maximization problems in communication networks.

Communication Networks with a Maximization Problem				
Reference	Framework	Constraints	Implementation	Budget Balance
[98]	Flow Control	System-wide	Partial	No
[63]	Flow Control	System-wide	Full	Yes
[99]	Flow Control	System-wide	Full	Yes
[91]	Flow Control	System-wide	Partial	Yes
[92]	Joint Flow Control & Multipath Routing	System-wide	Full	No
[100]	Power allocation & Spectrum sharing	System-wide	Full	Yes
[101]	Electric Vehicles	System-wide	Full	Yes
[102]	Networked Public Goods	Local	Full	Yes
[78]	Networked Public Goods	System-wide	Full	Yes
[77]	Network Utility Max	System-wide & Linear	Full	Yes

VCG-based Mechanisms for Communications Networks

"How can we allocate a fixed amount of an infinitely divisible resource among a finite set of strategic agents?" This is a classical problem in communication problems where pricing plays a key role [63], [103], [104]. Under natural network constraints, and if our desirable outcome is to achieve efficiency, then our starting point is the *Kelly mechanism* [105]. Briefly, the Kelly mechanism asks each agents to act as a bidder in an auction and announce a bid. Then the allocation of the single resource is conducted in proportion of the agents' bids. For each allocation, the social planner will receive an equal amount of tax from the agent's bid. In economics, we call this a "market-clearing price." Such a price has many desirable

characteristics as it ensures the bare-minimum communication (one-dimensional messages) and the social planner only needs to communicate back a single price per unit of resource. Both characteristics are ideal as they ensure the mechanism is practical, easily implementable for large-scale systems. The usefulness of the Kelly mechanism led to extensive research and the birth of *scalar-parameterized mechanisms* (as the only communication between agents and the social planner is a scalar). Mathematically, the allocation function for agent $i \in \mathcal{I}$ in the Kelly mechanism is given by $x_i = \frac{c \cdot p_i}{\sum_{j \in \mathcal{I} \setminus \{i\}} p_j}$, where $c \in \mathbb{N}$ is the capacity of a single resource.

Following a similar methodology, a special formulation of the VCG mechanism for an auction setting was introduced in [106], [107]. The problem was for a single divisible good under two different scenarios. At first, Semret [107] studied non-differentiable pseudo-utility functions between a social planner and multiple agents. In [106], the case was that agents seek bundles of links of a communication network (e.g., route) and the each agent's utility function depends on the minimum allocation received along their route. The key approach here was that for each link of the network a completely new and different auction was run. Both words have been somewhat generalized for multiple divisible links [108], [109]. A general convex VCG-based mechanism was introduced in [63] following [98] which just like in the Kelly mechanism only required one-dimensional bid signals from all agents. Following this work, Yang and Hajek [99] proposed a VCG-based mechanism with agents reporting a two-dimensional bid, i.e., a per-unit price β and a maximum quantity d_i that agent $i \in \mathcal{I}$ is willing to pay for the resource. Thus, the valuation function takes the form of

$$\hat{v}_i(x_i) = \beta \cdot \min\{x_i, d_i\}. \tag{14}$$

The benefit of this mechanism is it provides the equilibrium of the induced game (auction in this case) results to an optimal solution of the utility maximization problem based on the reported by the agents utility functions. Each agent pays for their allocation exactly the externality they impose on the other agents by participating in this mechanism. Of course, other mechanisms have been investigated, explored, and developed in great length [93]–[97].

Power Grid Systems

Over the last few decades, considerable efforts have been made to decarbonize our society's increasing energy demands [110] and reassess the efficiency of the existing methods of producing, managing, and consuming electricity [111]. Since 1980s [112]–[114], *demand-size management* (DSM) programs have been the standard way to study power-grid systems, their efficiency, and recently, their transformation to smart grid systems thanks to automation. Based on the US Energy Information Administration department: "Demand-side management (DSM) programs consist of the planning, implementing, and monitoring activities of electric utilities which are

the urban area a number of $c_{ij} \in \mathbb{N}$ times. Next, we have the following binary variables:

$$d_{ij}(t) = \begin{cases} 1, & \text{if } c_{ij} \text{ consists of time interval } t, \\ 0, & \text{otherwise.} \end{cases} \tag{26}$$

In addition, we have

$$x_{ij}(t) = \begin{cases} 1, & \text{if } c_{ij} \text{ is accepted,} \\ 0, & \text{otherwise.} \end{cases} \tag{27}$$

Thus, the optimization problem is formulated as follows:

$$\max_{x} \sum_{i \in \mathcal{I}} \sum_{j=1}^{m_i} p_{ij} \cdot x_{ij}, \tag{28}$$

$$\text{subject to: } \sum_{i \in \mathcal{I}} \sum_{j=1}^{m_i} d_{ij}(t) \cdot x_{ij}(t) \leq D_{\max}, \quad \forall t \in \mathcal{T}, \tag{29}$$

$$\sum_{j=1}^{m_i} x_{ij} \leq 1, \quad \forall i \in \mathcal{I}, \tag{30}$$

where $D_{\max}$ is the maximum number of vehicles that are allowed to be in the specific urban area. This optimization problem is rather difficult to solve as it is combinatorial and depending on its size, it can be almost impossible to solve in finite time. However, the auction-based congestion pricing [164] can provide good-enough solutions. In [167], a similar yet improved combinatorial optimization problem is proposed and using VCG-based incentives provides an efficient solution in the alleviation of traffic.

More recently, auctions and mechanism design have been used in ridesharing [5], [216] and autonomous vehicles public transportation problems [217]. We summarize some the latest papers in this area in Table 2.

Security Systems

It is well-known that cyber-defense remains a top priority for many organizations across different sectors. Part of this is because cyber-attacks are continuous and have the ability to bring down vital systems. Although there are many different angles to study security in a system of multiple agents, one approach is assume security is an economic good or resource and adopt game-theoretic approaches to study how we can find the best possible processes (via the appropriate design of incentives) for an efficient security investment [224], [225]. The main assumption in this approach is that the interactions between strategic agents in a system in which security is of crucial importance, can constitute a game. The objective of such a game is the provision of security in the means of investments by the agents. Thus, we say that security is

TABLE 2: A summary of auction-based mechanisms for transportation systems.

Auction-based mechanisms for transportation systems			
Reference	Model	Auction Type	Scenarios
[218]	Dynamic Ride-sharing	Combinatorial double auction	One-to-one assignment
[219]	Dynamic Ride-sharing	Vickrey	One-to-one assignment with detours
[220]	Dynamic Ride-sharing	Combinatorial double auction	One-to-one assignment with detours
[217]	Dial-a-Ride problem	VCG	One-to-many
[221]	Dial-a-Ride problem	Combinatorial double auction	One-to-many
[222], [223]	Dial-a-Ride problem	VCG	One-to-one assignment with detours

a public good. This approach was first introduced by [226] as a way for a "security game" to study airlines' baggage checking systems and what incentives are best. In parallel, Varian [227] studied security games for the reliability of computer systems. Afterward security games were extensively studied in different fields and applications [226]–[231]. There two surveys on this topic [232], [233].

The Design of Security Games

What is a security game? In general, interconnected systems of multiple agents who each depend on each other can be vulnerable to attacks by outsiders and external forces. Thus, any agent is encouraged to invest in the system's security measures not only to protect themselves but also to protect the other agents. Interconnectedness implies a positive externality. Consequently, the provision of security can be modeled naturally as the provision of a *public good* (non-rivalrous commodity) [1]. Thus, a security game models the strategic interactions of agents in a security system where each is asked to invest their own resources to secure the system from external attacks. Mathematically, we consider a network of $n \in \mathbb{N}$ agents (e.g., networked computer servers, corporate divisions, self-driving vehicles on a highway). The utilities of all agents are interdependent, and so $v_i(\mathbf{x})$. Each agent is assumed to possess a finite amount of resources $w_i \in \mathbb{N}$ available for investment, and we say that if an attack is successful against agent i's resources then a loss $\ell_i \in (0, w_i]$ may be imposed to agent $i \in \mathcal{I}$. The agent is allowed to protect their resources, hence agent $i \in \mathcal{I}$ may invest $x_i \in \mathbb{R}_{\geq 0}$. However, this investment comes with

a cost, represented by a general function $c_i : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ evaluated at the individual amount of investment x_i. We denote the vector of all agents' investments by $\mathbf{x} = \{x_1, \ldots, x_n\} = \mathcal{X}$. Then to capture how likely an attack may be successful or not, we introduce a risk function $r_i : \mathbb{R}^n_{\geq 0} \to [0, 1]$. Let $\mathbf{x}_{-i} = \{x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n\}$ capture the interdependence of all other agents' security. Then, the utility of an agent $i \in \mathcal{I}$ is of the form

$$v_i(\mathbf{x}) = w_i - \ell_i \cdot r_i(\mathbf{x}) - c_i(x_i). \tag{31}$$

Hence, the security game is given by the tuple $\langle \mathcal{I}, \mathcal{X}, (v_i)_{i \in \mathcal{I}} \rangle$ under a complete or incomplete information setting (depending on the problem).

The transformation of a security game to a mechanism design problem is straightforward. For a solution concept, we adopt the NE. Formally, we have

$$\tilde{x}_i = \arg\max_{x \geq 0} v_i(x_i, x_{-i}). \tag{32}$$

For completeness, the socially-optimal solution profile $\mathbf{x}^*$ can be computed as follows

$$\mathbf{x}^* = \arg\max_{x \geq 0} \sum_{i \in \mathcal{I}} v_j(\mathbf{x}). \tag{33}$$

Next, the total interdependent utility of agent $i \in \mathcal{I}$ is given by

$$u_i(\mathbf{x}, p_i) = v_i(\mathbf{x}) - p_i, \tag{34}$$

where p_i represents a payment/tax for agent $i \in \mathcal{I}$ when the investment profile is $\mathbf{x}$. Naturally, in a security system, it is imperative to design the right payments for all agents in order to incentivize voluntary participation. One approach for this is to introduce the notion of *exit equilibrium*: this is an equilibrium that takes into account the "external strategy" of an agent from the mechanism for the provision of a non-excludable goods, such as security and investment. At an exit equilibrium, an agent may unilaterally opt out of the mechanism and decide to adopt their best-response against the other agents who have chosen to participate in the mechanism [234]. Formally, we have

$$\hat{\mathbf{x}}^i_{-i} = \arg\max_{\mathbf{x}_{-i} \geq 0} \sum_{j \neq i} v_j(\mathbf{x}_{-\mathbf{i}}, \hat{\mathbf{x}}^i_i), \tag{35}$$

$$\hat{\mathbf{x}}^i_i = \arg\max_{x_i \geq 0} v_i(\hat{\mathbf{x}}^i_{-i}, x_i). \tag{36}$$

Most notably, although this is a clever notion of an exit equilibrium, Naghizadeh and Liu [234] showed an impossibility result for mechanisms that induce a security game with social optimality, weak budget balance, and voluntary participation. Most interestingly, two main approaches, i.e., the VCG mechanism and the *externality mechanism* [235], [236] that have been quite successful in other applications fail to circumvent this theoretical obstacle. For the purpose of this article, we focus on the latter only.

The externality mechanism aims to redistribute the wealth that collects from the agents and ensure strong budget balance (all payments equal to zero). As it was shown in [236], this mechanisms induces a game with social optimality, voluntary participation, and maintains a balanced budget for cellular networks (which is an excludable public good). The payment function

$$t_i(\mathbf{x}^*) = -\sum_{j\in\mathbb{I}} x_j^* \ell_i \frac{\partial r_i}{\partial x_j}(\mathbf{x}^*) - x_i^* \frac{\partial c_i}{\partial x_i}(x_i^*). \tag{37}$$

These taxes fail to induce voluntary participation for all agents, and thus the system will be significantly vulnerable to attacks as its security cannot be guaranteed.

Security in Communication Systems

TABLE 3: A snapshot of papers on security of communication systems.

Auction-based mechanisms for security in communication systems.			
Auction Type	Characteristics	Solution Concept	Scenario
English auction [237]	Winner pays second highest price	NE	Information integrity
Dutch auction [237]	Winner pays final price	NE	Black-hole attack
First-price sealed-bid auction [238]	Winner pays highest price	NE	Privacy or faked-sensing attack
Vickrey auction [238]	Winner pays second highest price	NE	User collusion/bid-rigging
VCG auction [239]	Vickrey auction with multiple goods	BNE	Eavesdropping attack
Share auction [240]	Matching of goods based on ratio of buyers' bids	NE	Distributed Denial-of-Service attack
Ascending-clock auction [241]	Increase price until demand equals supply	Walrasian equilibrium	False-name bids
Double auction [242]	Matching between sellers' demands and buyers' bids	Market equilibrium	Privacy

Security has been extensively studied in communication systems since the 1980s. With the advancement of communication technology such as wireless networks, communication between

systems, computers, phones has never been as easy and quick. However, such systems admit vulnerabilities and since they play a key role in the exchange of information, attacks are continuous and frequent. So, in this subsection, we briefly focus on auction-based mechanisms that offer solutions for an array of communication scenarios and the solutions we can derive.

Auctions have been used to model the economic-in-nature interactions between agents of a communication system that need to invest resources for their security and wellbeing of the system (e.g., prevent external attacks). Examples of auctions used to model security include *English auctions*, *Dutch auctions*, *Sealed-bid auctions*, *Double auctions*, *Share auctions*, and *Ascending-clock auctions*.

Both English and Dutch auctions are multiple-round auctions and allow buyers to exchange information (their bid) to other buyers. In economics, this type of disclosure in auctions is called *open-outcry*. In addition, the auctioneer in English auctions introduces a bid and keeps increasing it at each round, while in Dutch auctions, the opposite process occurs (high price decreasing until accepted). Thanks to the simplicity of the auction rules, such auctions have been used to defend and identify "malicious" agents in a system [243]. Sealed-bid auctions, as their name suggest, are characterized by their ensured privacy as buyers do not disclose their bids to anyone (examples are the first-price auction, Vickrey auction, and the VCG auction). The specifics of each auction have been summarized in Table 3 together with recent key applications [244]. For example, the VCG auction extends the Vickrey auction with multiple goods and sets the right rules to determine the winner (i.e., the buyer who gets the auctioned good). In particular, the VCG auction awards the good to the buyer, who then pays the second-highest bid. This is of extreme importance as it establishes the following principle: bid your true valuation and pay less than you expect. The VCG auction has been widely used in securing communication and wireless networks (e.g., to prevent agent misbehavior).

So far, the above-mentioned auctions are *one-sided*, meaning there is one auctioneer who tries to sell one or more goods to a finite number of buyers. In a double auction there is still an auctioneer who manages an auction between multiple buyers, who each submit bids at the same time, and sellers, who each submit payment demands for their goods [242]. For example, an auctioneer can match the buyers and sellers as follows: list the bids in a descending order and the payment demands in an ascending order. Denote by p_m^{seller}, p_m^{buyer} for the payment demand of a seller and the bid of a buyer, respectively. Then the auctioneer can find the largest index $m \in \mathbb{N}$ for which we have

$$p_m^{\text{buyer}} \leq p_m^{\text{seller}}. \tag{38}$$